



OPEN Application of novel similarity measures in electric vehicle charging station site selection based on q-rung orthopair hesitant fuzzy rough sets under quantitative information

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Similarity measures provide a viable approach for evaluating the degree of similarity between two collections. They serve as effective procedures for addressing decision-making issues influenced by cognitive processes. This manuscript presents the innovative notions of cosine similarity and weighted cosine similarity measures based on q-rung orthopair hesitant fuzzy rough sets and discusses some relevant properties of the aforementioned measures. These measures are an enhancement of several existing ones that can handle ambiguous and imprecise information with a broader range. Decision makers can determine the most desirable alternative by evaluating the differences between each option and the portrayed standard. Moreover, we present an exemplary illustration for the assessment of an optimal location for electric vehicle charging station in order to illustrate the applicability and advantages of the established similarity measures. Furthermore, assess and determine whether a geographical area has a good, moderate or hazardous air quality index (AQI). The AQI is one of the primary aspects of the environment which is affected by air pollution. Air pollution is one of the extensive worldwide problems, and currently it is well accepted to be harmful to human health. To emphasise the authenticity and the viability of the suggested approach, the findings obtained through cosine and weighted cosine similarity measures are compared. It is pertinent to observe that the computation from novel similarity measures is consistent and provides the same ranking order. We validate the efficacy of the proposed similarity measures through a comparative analysis with established approaches, contrasting their rankings with those demonstrated in existing literature. The results illustrate that the novel similarity measures produce identical ranking patterns and are capable of discerning differences between patterns. Furthermore, the findings demonstrate that the proposed measures are not restricted to a specific domain, but can be effectively utilized in a wide variety of applications.

Keywords Similarity measures, Q-rung orthopair hesitant fuzzy rough sets, Electric vehicle charging station, Air quality index

Similarity measures (SMs) are essential tools for evaluating the degree of similarity between two distinct items and represent a crucial component of fuzzy sets (FSs) theory, with widespread applications in pattern recognition, medical diagnosis, and numerous other fields. In numerous fields, that involve data mining, artificial intelligence, and information recollection, the assessment of similarity is crucial for facilitating informed decision-making (DM). Similarity measures provide a numerical evaluation of the degree of resemblance association, or closeness

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between entities, enabling informed DM across a range of applications, such as clustering, which involves the identification of patterns or structures by grouping similar objects or entities, thereby facilitating intelligent choices related to data segmentation, customer profiling, or resource allocation. The classification entails the systematic allocation of labels or categories to entities through the assessment of their resemblance to predefined instances, thereby facilitating informed decisions in areas such as content moderation, sentiment analysis, or fraud detection. The growing significance of similarity measures originates from the rapid expansion in data volume, diversity, and complexity. As data proliferates and grows increasingly complex, it is imperative to cultivate effective, precise, and resilient measures of similarity for proficient DM that oversee large data sets and enable swift analysis. The reliable decision-making ensures accuracy and minimizing errors in the DM processes. The enhancement of similarity measures has the potential to unveil novel opportunities in data analysis, machine learning, and information retrieval, thereby fostering advancement across diverse domains and applications while also facilitating effective DM. Researchers are interested in measures of similarity between FSs because of their wide range of applications. Li and Cheng¹ investigated SMs based on intuitionistic fuzzy sets (IFSs) and applied them to pattern recognition problems. Subsequently, Liang and Shi² presented various similarity measures for IFSs and evaluated their implications in DM challenges. Szmidt and Kacprzyk^{3,4} introduced SMs between IFSs utilising distance and its applications to DM challenges. Hung and Yang⁵ introduced a technique utilising the Hausdorff distance to identify various similarity measures suitable for linguistic variables among IFSs. Afterwards, Liu⁶ introduced several SMs that establish interactions between IFSs and employed them for pattern recognition. Xu⁷ developed a collection of IFS-based SMs and applied them to the DM problem through the use of IF information. Gogoi et al.⁸ investigated the statistical model for interval-valued fuzzy numbers and its relevance to risk assessments in rice cultivation. Xu and Chen⁹ proposed a collection of SMs and distance measures that represent various combinations and extensions of the weighted Hamming, Euclidean, and Hausdorff distances. Wei et al.¹⁰ introduced a generalised entropy measure applicable to IFSs and Pythagorean fuzzy sets (PyFSs). In addition, a method has been established for the construction of SM for IFS and PyFSs through the application of entropy measures. Chachi et al.¹¹ developed a comprehensive approach for assessing the similarity among IFSs. A wide range of research investigations (refer to^{12–14} for further details) have explored a range of information measures, comprising distance measures, inclusion measures, and entropy measures in relation to IFSs and PyFSs, as well as discussed their fundamental relationships. Hussain and Yang¹⁵ conducted an investigation of the distance and SMs of PyFSs utilising the Hausdorff measures, and then applied their results using fuzzy TOPSIS approach. Rahim et al.¹⁶ conducted an investigation through various cosine SMs and distance measures based on (p, q) -quasirung orthopair FSs, and elaborated their applications within the realm of investment DM. Patel et al.¹⁷ examined the similarity measure based on intuitionistic fuzzy sets and investigated its applicability in pattern recognition and the integration of multimodal medical images. This method's usefulness was proved by extensive trials using three different kinds of medical imaging datasets. Rani et al.¹⁸ established a novel similarity measure concerning picture FSs. The suggested SM among PFs has the potential to address the limitations present in existing SMs between PFs. A innovative DM approach has been formulated, based on the suggested similarity measure and the measurement of alternatives and ranking according to the compromise solution (MARCOS) technique is established for DM. Zeeshan et al.¹⁹ established the novel notion of complex Pythagorean fuzzy soft matrices (CPFMSs) and developed a series of new distance metrics between them. Furthermore, they described a real-world application based on numerical examples, and comparison analysis had been appropriately demonstrated. A variety of studies have concentrated on the information measures related to q -rung orthopair FSs (q -ROFSs). To address the issue of comparability, Peng et al.²⁰ introduced an innovative score function for q -ROF numbers. Thereafter, they formulated a distance measure for q -ROFSs characterized by diverse parameters and substantiated their findings with appropriate rationale. Moreover, the various beneficial characteristics of the resulting distance and SM are determined. Muthukumar et al.²¹ investigated the use of similarity measurements for intuitionistic fuzzy soft sets in medical diagnosis. Peng and Liu²² explored the applications of SMs distance, entropy, and inclusion measures in the context of q -ROFSs. Dinesh and Kumar²³ presented knowledge measure and SMs along with a numerical implementation, to evaluate the viability of the suggested technique. Wang et al.²⁴ have formulated a collection of SMs and distance measures using Fermatean fuzzy sets. Their analysis of pattern recognition, medical diagnostics, and DM stipulates considerable evidence and importance regarding the measures discussed. Gupta and Kumar²⁵ developed innovative similarity measures based in hesitant FSs, implemented the developed measures in the fields of pattern recognition and clustering analysis, and assessed their efficiency index. Hussain et al.²⁶ introduced similarity measures for clustering that employ Pythagorean FSs together with the Jaccard index. Ali and Naeem²⁷ developed cosine similarity measures in the context of q -rung orthopair linguistic sets and demonstrated their applicability via numerical example for the DM problem. Vishnukumar et al.²⁸ established a new SMs using a unique accuracy function for interval-valued intuitionistic fuzzy numbers and discussed their utilization in real-world DM processes. Jafar et al.²⁹ suggested the use of trigonometric similarity measures, specifically cosine and cotangent, along with weighted trigonometric similarity measures, to address the complexities associated with the selection of renewable energy sources, and expressed the research's significance in practical real-world challenges.

Pawlak³⁰ introduced the framework for rough sets (RSs) theory, captivated the interest of researchers and scientists across various domains, including financial analysis, image processing, information retrieval, medical diagnosis, data mining, and feature selection. Cornelis et al.³¹ initialed the notion of intuitionistic fuzzy RSs (IFRSs) through the application of lower and upper approximations of IFSs. Recently, Attaullah et al.³² broadened the idea of fuzzy RSs and presented the innovative idea of q -rung orthopair hesitant fuzzy RSs (q -ROHFRSs), which represents a hybrid structure of RSs and q -ROHFSs, and explored its application in DM problems. A cosine similarity measure based on Bhattacharya's distance³³ is among the numerous similarity measures. The concept involves the inner product of two vectors, which is then divided by the product of their magnitudes. The cosine similarity performs as a prominent tool for evaluating vector similarity, commonly employed in

the domain of information retrieval. Zhang et al.³⁴ investigated various prospective of fuzzy rough set models in the framework of multi-source complex information systems utilizing SMs. Attaullah et al.³⁵ formulated innovative cosine and weighted cosine SMs based on intuitionistic hesitant fuzzy rough sets and examined their fundamental properties. An illustrative scenario from the assessment of volatile currency in Pakistan is presented to demonstrate the validity of the proposed SMs and confirmed the effectiveness of the suggested technique. The findings indicate that the existing SMs exhibit specific constraints, including outcomes that lack justification, the issue of a zero denominator, results that challenge intelligence, and flouting the similarity axioms. The fundamental shortcomings present in the existing methodologies can adversely impact the results of decision-making processes. Therefore, it is essential to establish a novel effective similarity measure to overcome these constraints. This study introduces a novel SMs and illustrates its superiority through a comparative analysis with established methods. Moreover, a variety of pattern recognition methodologies have been developed employing multiple SMs. Certain methodologies demonstrated limitations in accurately categorizing unknown patterns across particular circumstances, and some displayed a lack of reliability in pattern recognition owing to their exceedingly low confidence levels. This study introduces a methodology for pattern recognition that employs innovative SMs in the framework of q -ROHFRS. However, conventional cosine SMs are insufficient for determining cosine similarity between conventional fuzzy sets that exhibit hesitancy, inaccuracy, and incapable of tackle the intricate real world challenges. The goals of this research are follows:

1. Establish the improved cosine SM and weighted cosine SMs based on q -ROHFRS for calculating the degree of similarity between q -ROHFRS by extending the existing SM based on the cosine SM³³. This enhancement promotes more dependable DM in scenarios characterised by multiple criteria.
2. Analyse their related characteristics comprehensively.
3. Implement novel SMs in identifying the optimal site for an electric vehicle charging station and assess and determine whether a geographical area has a good, moderate or hazardous air quality index (AQI) to accomplish the applicability of the established SMs. The illustrated example demonstrates the application of the suggested approach within decision-making scenarios.
4. The outcomes obtained from the suggested SMs are compared with those established in the recent literature. The analysis indicates that the outcomes of the novel SMs are more effective, rational, and maintain the same ranking order, and useful for resolving the practically ambiguous environment.

The remaining structure of the article is as follows: “**Basic concepts**” section includes some essential concepts of FS, HFS, IHFS, and q -ROFS, RS, and q -ROHFRS. In “**Similarity measures**” section, we establish the cosine similarity functions. The subsequent “**Practical applications**” section, initiates some new types of cosine and weighted cosine SMs for q -ROHFRS and explore their basic properties. “**Validation of the proposed approach: a comparative study**” section demonstrates the applications of SMs between q -ROHFRSs for the identification of an optimal location for electric vehicle charging station. “**Conclusions and future scopes**” section summarises the findings of the investigations and future recommendations. The organisation of the proposed work is depicted in Fig. 1.

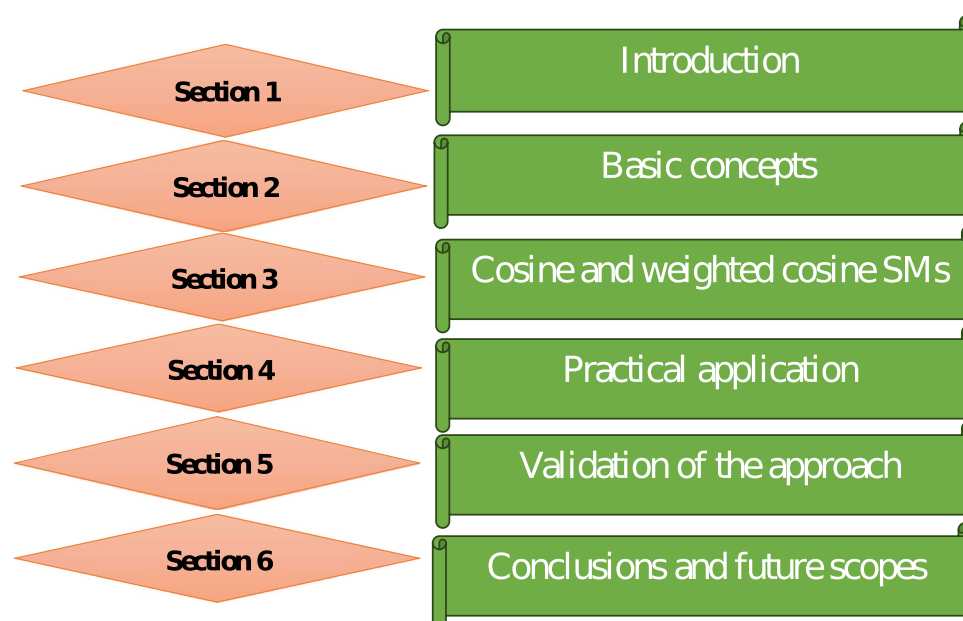


Fig. 1. The graphic representation of the structure of the paper.

Basic concepts

Herein, we highlight the basic notations along with the corresponding operating rules. The readers will be able to better understand the key concepts of intended investigation.

Definition 2.1 ³⁶ $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$ be a universal set. By q -ROHFS $\mathcal{Q}$ over ϱ , we mean a structure define as:

$$\mathcal{Q} = \{\langle \varsigma_i, \mathcal{B}_{h_{\mathcal{Q}}}(\varsigma_i), \mathcal{U}_{h_{\mathcal{Q}}}(\varsigma_i) \rangle \mid \varsigma_i \in \varrho\},$$

where $\mathcal{B}_{h_{\mathcal{Q}}}(\varsigma_i)$ is membership grade (MG) and $\mathcal{U}_{h_{\mathcal{Q}}}(\varsigma_i)$ is non-membership grade (NMG) with the property that for all $\varsigma_i \in \varrho$, $\forall \mu_{\mathcal{Q}} \in \mathcal{B}_{h_{\mathcal{Q}}}(\varsigma_i)$, $\forall \nu_{\mathcal{Q}}(\varsigma_i) \in \mathcal{U}_{h_{\mathcal{Q}}}(\varsigma_i)$ such that $(\max(\mu_{\mathcal{Q}}(\varsigma_i)))^q + (\min(\nu_{\mathcal{Q}}(\varsigma_i)))^q \leq 1$.

Definition 2.2 ³⁰ For a universal set $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$, $\mathfrak{S}$ be any relation on ϱ . The mapping $\mathfrak{S}^* : \varrho \rightarrow M(\varrho)$ by $\mathfrak{S}^*(\varsigma_i) = \{a \in \varrho \mid (\varsigma_i, a) \in \mathfrak{S}\}$, for $\varsigma_i \in \mathcal{P}$ where $\mathfrak{S}^*(\varsigma_i)$ is called a successor neighborhood of the element ς_i with respect to relation $\mathfrak{S}$. The pair $(\varrho, \mathfrak{S})$ is known as (crisp) approximation space. The lower approximation (LA) and upper approximation (UA) for any set $\mathcal{T} \subseteq \varrho$ with respect to $(\varrho, \mathfrak{S})$ is outlined as:

$$\underline{\mathfrak{S}}(\mathcal{T}) = \{\varsigma_i \in \varrho \mid \mathfrak{S}^*(\varsigma_i) \subseteq \mathcal{T}\}; \quad \overline{\mathfrak{S}}(\mathcal{T}) = \{\varsigma_i \in \varrho \mid \mathfrak{S}^*(\varsigma_i) \cap \mathcal{T} \neq \Phi\}.$$

The pair $(\underline{\mathfrak{S}}(\mathcal{T}), \overline{\mathfrak{S}}(\mathcal{T}))$ is called RS.

Definition 2.3 ³⁰ Let $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$ be a fixed set and $\mathfrak{S} \in IFS(\varrho \times \varrho)$ be an IF relation. Then

- (1) $\mathfrak{S}$ is reflexive if $\mu_{\mathfrak{S}}(\varsigma_i, \varsigma_i) = 1$ and $\nu_{\mathfrak{S}}(\varsigma_i, \varsigma_i) = 0$, $\forall \varsigma_i \in \varrho$;
- (2) $\mathfrak{S}$ is symmetric if $\forall (\varsigma_i, a) \in \varrho \times \varrho$, $\mu_{\mathfrak{S}}(\varsigma_i, a) = \mu_{\mathfrak{S}}(a, \varsigma_i)$ and $\nu_{\mathfrak{S}}(\varsigma_i, a) = \nu_{\mathfrak{S}}(a, \varsigma_i)$;
- (3) $\mathfrak{S}$ is transitive if $\forall (\varsigma_i, b) \in \varrho \times \varrho$,

$$\mu_{\mathfrak{S}}(\varsigma_i, b) \geq \bigvee_{a \in \varrho} [\mu_{\mathfrak{S}}(\varsigma_i, a) \wedge \mu_{\mathfrak{S}}(a, b)]; \quad \text{and} \quad \nu_{\mathfrak{S}}(\varsigma_i, b) = \bigwedge_{a \in \varrho} [\nu_{\mathfrak{S}}(\varsigma_i, a) \wedge \nu_{\mathfrak{S}}(a, b)].$$

Definition 2.4 Let $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$ be a fixed set. Then any $\mathfrak{S} \in q - RFS(\varrho \times \varrho)$ is called q -rung relation. The pair $(\varrho, \mathfrak{S})$ is said to be q -rung approximation space. For any $\mathcal{T} \subseteq q - RFS(\varrho)$, the LA and UA of $\mathcal{T}$ with respect to q -RF space $(\varrho, \mathfrak{S})$ are two q -RFSS, represented by $\underline{\mathfrak{S}}(\mathcal{T})$ and $\overline{\mathfrak{S}}(\mathcal{T})$ and outlined by:

$$\overline{\mathfrak{S}}(\mathcal{T}) = \{\langle \varsigma_i, \mu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) \rangle \mid \varsigma_i \in \varrho\}; \quad \underline{\mathfrak{S}}(\mathcal{T}) = \{\langle \varsigma_i, \mu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) \rangle \mid \varsigma_i \in \varrho\};$$

where

$$\begin{aligned} \mu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) &= \bigvee_{g \in \mathcal{P}} [\mu_{\mathfrak{S}}(\varsigma_i, g) \vee \mu_{\mathcal{T}}(g)]; & \nu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) &= \bigwedge_{g \in \varrho} [\nu_{\mathfrak{S}}(\varsigma_i, c) \wedge \nu_{\mathcal{T}}(g)]; \\ \mu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) &= \bigwedge_{g \in \mathcal{P}} [\mu_{\mathfrak{S}}(\varsigma_i, c) \wedge \mu_{\mathcal{T}}(g)]; & \nu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i) &= \bigvee_{g \in \varrho} [\nu_{\mathfrak{S}}(\varsigma_i, c) \vee \nu_{\mathcal{T}}(g)]; \end{aligned}$$

such that

$$0 \leq ((\mu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i))^q + (\nu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i))^q) \leq 1, \quad \text{and} \quad 0 \leq ((\mu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i))^q + (\nu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i))^q) \leq 1.$$

As $(\underline{\mathfrak{S}}(\mathcal{T}), \overline{\mathfrak{S}}(\mathcal{T}))$ are $q - RFS$ s, so $\underline{\mathfrak{S}}(\mathcal{T}), \overline{\mathfrak{S}}(\mathcal{T}) : q - RFS(\varrho) \rightarrow q - RFS(\varrho)$ are LA and UA operators.

The pair

$$\mathfrak{S}(\mathcal{T}) = (\underline{\mathfrak{S}}(\mathcal{T}), \overline{\mathfrak{S}}(\mathcal{T})) = \{\langle \varsigma_i, (\mu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i)), (\mu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i)) \rangle \mid \varsigma_i \in \mathcal{T}\}$$

is known as q -RFRS. For simplicity

$$\mathfrak{S}(\mathcal{T}) = \{\langle \varsigma_i, \mu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\underline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), (\mu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i), \nu_{\overline{\mathfrak{S}}(\mathcal{T})}(\varsigma_i)) \rangle \mid \varsigma_i \in \varrho\}$$

is represented as $\mathfrak{S}(\mathcal{T}) = ((\underline{\mu}, \underline{\nu}), (\overline{\mu}, \overline{\nu}))$ and is known as q -RFRV.

Definition 2.5 ³² Let $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$ be a fixed set and for any subset $\mathfrak{S} \in q - ROHFS(\varrho \times \varrho)$ is said to be q -rung HF relation. The pair $(\varrho, \mathfrak{S})$ is said to be q -ROHF approximation space. If for any $\mathcal{T} \subseteq q - ROHFS(\varrho)$, then the LA and UA approximations of $\mathcal{T}$ with respect to q -ROHF approximation space $(\varrho, \mathfrak{S})$ are two q -ROHFSs, represented by $\underline{\mathfrak{S}}(\mathcal{T})$ and $\overline{\mathfrak{S}}(\mathcal{T})$ and defined by:

$$\overline{\mathfrak{S}}(\mathcal{T}) = \{\langle \varsigma_i, \mathcal{B}_{h_{\overline{\mathfrak{S}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\overline{\mathfrak{S}}(\mathcal{T})}}(\varsigma_i) \rangle \mid \varsigma_i \in \varrho\}; \quad \underline{\mathfrak{S}}(\mathcal{T}) = \{\langle \varsigma_i, \mathcal{B}_{h_{\underline{\mathfrak{S}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\underline{\mathfrak{S}}(\mathcal{T})}}(\varsigma_i) \rangle \mid \varsigma_i \in \varrho\};$$

where

$$\begin{aligned}\mathcal{B}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) &= \bigvee_{k \in \varrho} [\mathcal{B}_{h_{\mathfrak{T}}}(\varsigma_i, k) \vee \mathcal{B}_{h_{\mathcal{T}}}(k)]; & \mathcal{U}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) &= \bigwedge_{k \in \varrho} [\mathcal{U}_{h_{\mathfrak{T}}}(\varsigma_i, k) \wedge \mathcal{U}_{h_{\mathcal{T}}}(k)]; \\ \mathcal{B}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) &= \bigwedge_{k \in \varrho} [\mathcal{B}_{h_{\mathfrak{T}}}(\varsigma_i, k) \wedge \mathcal{B}_{h_{\mathcal{T}}}(k)]; & \mathcal{U}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) &= \bigvee_{k \in \varrho} [\mathcal{U}_{h_{\mathfrak{T}}}(\varsigma_i, k) \vee \mathcal{U}_{h_{\mathcal{T}}}(k)];\end{aligned}$$

such that $0 \leq \left(\max(\mathcal{B}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i)) \right)^q + \left(\min(\mathcal{U}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i)) \right)^q \leq 1$ and $0 \leq \left(\min(\mathcal{B}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i)) \right)^q + \left(\max(\mathcal{U}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i)) \right)^q \leq 1$. As $(\overline{\mathfrak{T}}(\mathcal{T}), \underline{\mathfrak{T}}(\mathcal{T}))$ are q -ROHFSs, so $\overline{\mathfrak{T}}(\mathcal{T}), \underline{\mathfrak{T}}(\mathcal{T}) : q\text{-ROHFS}(\varrho) \rightarrow q\text{-RFS}(\varrho)$ are LA and UA approximation operators. The pair

$$\mathfrak{T}(\mathcal{T}) = (\underline{\mathfrak{T}}(\mathcal{T}), \overline{\mathfrak{T}}(\mathcal{T})) = \left\{ \left\langle \varsigma_i, \left(\mathcal{B}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) \right), \left(\mathcal{B}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \mathcal{T} \right\}$$

will be called q -ROHFRS. For simplicity

$$\mathfrak{T}(\mathcal{T}) = \left\{ \left\langle \varsigma_i, \left(\mathcal{B}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\underline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) \right), \left(\mathcal{B}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i), \mathcal{U}_{h_{\overline{\mathfrak{T}}(\mathcal{T})}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \mathcal{T} \right\}$$

is represented by $\mathfrak{T}(\mathcal{T}) = ((\underline{\mathcal{B}}, \underline{\mathcal{U}}), (\overline{\mathcal{B}}, \overline{\mathcal{U}}))$ and is known as q -ROHFRV.

Similarity measures

Similarity measures are quantitative tools employed to evaluate the resemblance between datasets and are vital across multiple fields, including ecology, information retrieval, and pattern recognition and clinical diagnostics. In various domains of fuzzy sets, scholars have developed certain metrics to tackle practical issues. Ganie et al.¹⁴ introduced a novel picture fuzzy similarity measure and explored its practical applications through comprehensive benchmarking against existing metrics. Their findings confirmed its efficacy in both classification tasks and diagnostic analyses. Singh et al.³⁷ established similarity measures utilizing picture fuzzy sets (PIFSs) to evaluate flood hazards in southern India. Wei³⁸ introduced similarity metrics based on the cosine function for multi-attribute decision-making (MADM). Wei³⁹ introduced grey and set-theoretic similarity measures, which were utilized in mineral field identification and material classification. Thao⁴⁰ employed entropy-based similarity measures for MADM, whereas Khan et al.⁴¹ formulated bi-parametric Performance Index Function (PIF) metrics for medical diagnostics and classification. Additional advancements include generalized Dice metrics for material identification⁴², exponential function-based evaluation measures⁴³, and methods focused on membership levels⁴⁴. Further research^{44–47} explores hybrid methodologies.

Although existing similarity metrics carry advantages, but reveal significant shortcomings in handling ambiguous data and demonstrate an over-sensitivity to outliers, thereby undermining their reliability in intricate scenarios. This gap highlights the necessity for more rigorous and intellectually rational alternatives. This research addresses notable shortcomings in existing methodologies for assessing similarity measures as follows:

- *Axiomatic deficiencies* Numerous existing similarity measures are inconsistent with regard to essential foundational axiomatic criteria, thereby constraining their theoretical trustworthiness.
- *Inconsistent outcomes* When utilized in comparisons, existing metrics frequently produce unconventional or inconsistent outcomes, thereby compromising practical reliability.
- *Pattern recognition limitations* Existing methodologies exhibit insufficient distinguishing capability to discern unknown patterns within classification tasks, thereby diminishing their effectiveness in practical applications.

These limitations highlight the importance for a more robust, axiomatic, and versatile similarity measure to address the previously mentioned shortcomings. In view of these considerations, this paper introduces two main advancements: cosine and weighted cosine similarity measures based on q -ROHFRSs, along with a comprehensive decision-making framework.

The Cosine and weighted cosine similarity measures

The cosine SM is a fundamental measure of similarity between two n -dimensional vectors, using the cosine of the angle that separates them. The similarity of two vectors are determined by their directional alignment, with distance being neglected. The cosine similarity that exists between two attribute vectors $\mathcal{W} = \{\vartheta_1, \vartheta_2, \vartheta_3, \dots, \vartheta_n\}$ and $\mathcal{X} = \{\varpi_1, \varpi_2, \varpi_3, \dots, \varpi_n\}$ is expressed as follows:

$$\cos\theta = \frac{\bigoplus_{r=1}^n \vartheta_r \varpi_r}{\sqrt{\bigoplus_{r=1}^n \vartheta_r^2} \sqrt{\bigoplus_{r=1}^n \varpi_r^2}}.$$

Cosine similarity between two FSs $\mathcal{B}_{\mathcal{W}}(\vartheta_r)$ and $\mathcal{B}_{\mathcal{X}}(\vartheta_r)$, based on Bhattacharya's distance³³ is defined as follows:

$$C_{FS}(\mathcal{W}, \mathcal{X}) = \frac{\bigoplus_{r=1}^n \mathcal{B}_{\mathcal{W}}(\vartheta_r) \mathcal{B}_{\mathcal{X}}(\vartheta_r)}{\sqrt{\bigoplus_{r=1}^n (\mathcal{B}_{\mathcal{W}}(\vartheta_r))^2} \sqrt{\bigoplus_{r=1}^n (\mathcal{B}_{\mathcal{X}}(\vartheta_r))^2}}.$$

The technique used for describing the cosine similarity between two IFs in a two-dimensional vector space, as suggested by Ye⁴⁸, is outlined as follows:

$$C_{IFS}(\mathcal{W}, \mathcal{X}) = \frac{\oplus_{r=1}^n [\mathcal{B}_{\mathcal{W}}(\vartheta_r) \mathcal{B}_{\mathcal{X}}(\vartheta_r) + \mathcal{U}_{\mathcal{W}}(\vartheta_r) \mathcal{U}_{\mathcal{X}}(\vartheta_r)]}{\sqrt{\oplus_{r=1}^n (\mathcal{B}_{\mathcal{W}}(\vartheta_r))^2 + (\mathcal{U}_{\mathcal{W}}(\vartheta_r))^2} \sqrt{\oplus_{r=1}^n (\mathcal{B}_{\mathcal{X}}(\vartheta_r))^2 + (\mathcal{U}_{\mathcal{X}}(\vartheta_r))^2}},$$

where all vectors lies in $[0, 1]$.

Definition 3.1 *Cosine similarity measure for q-ROHFRSs* The cosine similarity of two vectors is determined by dividing the inner product of the vectors by the product of their magnitudes. The cosine SM serves as crucial measuring tool in the DM process, facilitating the evaluation of similarity between two vectors. Through the computation of the cosine of the angle that separates two vectors, decision makers are able to ascertain the extent of similarity that exists between them, with values spanning from 0 to 1. The advantages of cosine similarity in data mining are numerous, including efficient computation and the ability to quickly analyze large data sets, which facilitates timely DM. It also minimizes the influence of deviations and errors on the DM process, emphasizing the direction and relevance of data over its magnitude. Through an exhaustive understanding of the cosine SM, along with its various applications, benefits, and constraints, decision makers can adeptly utilize this technique to their advantage. In the realm of DM, cosine similarity serves as a formidable instrument for the analysis and interpretation of data, facilitating well-informed choices that propel success. The expression refers to the cosine of the angle produced by the intersection of two q-ROHFRS.

Let $\mathcal{W} = \left\{ \left\langle \varsigma_i, \left(\underline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i), \underline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \right), \left(\overline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i), \overline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \varrho \right\}$ and $\mathcal{X} = \left\{ \left\langle \varsigma_i, \left(\underline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i), \underline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \right), \left(\overline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i), \overline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \varrho \right\}$ be in $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$. Then the cosine SM between two q-ROHFRSs $\mathcal{W}$ and $\mathcal{X}$ are defined by:

$$C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = \frac{1}{n} \bigoplus_{r=1}^n \frac{\xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i) \xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i) + \xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i) \xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i)}{\sqrt{(\xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i))^2 + (\xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i))^2} \sqrt{(\xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i))^2 + (\xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i))^2}},$$

where

$$\begin{aligned} \xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i) &= \frac{\frac{1}{\mathcal{B}^*} \oplus_{r=1}^{\mathcal{B}^*} (\underline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i)) + \frac{1}{\mathcal{B}^*} \oplus_{r=1}^{\mathcal{B}^*} (\overline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i))}{2}, & \xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i) &= \frac{\frac{1}{\mathcal{B}^*} \oplus_{r=1}^{\mathcal{B}^*} (\underline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i)) + \frac{1}{\mathcal{B}^*} \oplus_{r=1}^{\mathcal{B}^*} (\overline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i))}{2}, \\ \xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i) &= \frac{\frac{1}{\mathcal{U}^*} \oplus_{r=1}^{\mathcal{U}^*} (\underline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i)) + \frac{1}{\mathcal{U}^*} \oplus_{r=1}^{\mathcal{U}^*} (\overline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i))}{2}, & \xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i) &= \frac{\frac{1}{\mathcal{U}^*} \oplus_{r=1}^{\mathcal{U}^*} (\underline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i)) + \frac{1}{\mathcal{U}^*} \oplus_{r=1}^{\mathcal{U}^*} (\overline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i))}{2}. \end{aligned}$$

Proposition 3.2 Let $\mathcal{W}$, $\mathcal{X}$ and $\mathcal{Y}$ be three q-ROHFRSs on ϱ . The cosine SM $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X})$ is a mapping $S : q-ROHFRSs(\varrho) \times q-ROHFRSs(\varrho) \rightarrow [0, 1]$, carrying the following properties:

- (A₁) $0 \leq C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) \leq 1$
- (A₂) $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = C_{q-ROHFRSs}(\mathcal{X}, \mathcal{W})$
- (A₃) $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = 1$ iff $\mathcal{W} = \mathcal{X}$
- (A₄) If $\mathcal{W} \subseteq \mathcal{X} \subseteq \mathcal{Y}$, then $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X})$ and

$$C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{X}, \mathcal{Y}).$$

Proof

(A₁): It is evident that all positive values of the cosine function are located within the interval of 0 to 1.

(A₂): The proof is straightforward.

(A₃): When $\mathcal{W} = \mathcal{X}$, then $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X})$ must be equal to 1. Conversely, if

$C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = 1$, then $\xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i) = \xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i)$ and $\xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i) = \xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i)$ that is,

$\underline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i) = \underline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i)$, $\overline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i) = \overline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i)$, $\underline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) = \underline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i)$, and $\overline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) = \overline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i)$. This implies that $\mathcal{W} = \mathcal{X}$. (A₄): If $\mathcal{W} \subseteq \mathcal{X} \subseteq \mathcal{Y}$, then it

follows that $\underline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i) \subseteq \underline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i) \subseteq \underline{\mathcal{B}}_{h_{\mathcal{Y}}}(\varsigma_i)$, $\overline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i) \subseteq \overline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i) \subseteq \overline{\mathcal{B}}_{h_{\mathcal{Y}}}(\varsigma_i)$ and

$\underline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \subseteq \underline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \subseteq \underline{\mathcal{U}}_{h_{\mathcal{Y}}}(\varsigma_i)$, $\overline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \subseteq \overline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \subseteq \overline{\mathcal{U}}_{h_{\mathcal{Y}}}(\varsigma_i)$.

In the interval $\left[0, \frac{\pi}{2}\right]$, the cosine function is a decreasing function. Therefore, $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X})$ and $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{X}, \mathcal{Y})$. $\square$

Definition 3.3 *Weighted cosine similarity measure for q-ROHFRSs*

Let $\mathcal{W} = \left\{ \left\langle \varsigma_i, \left(\underline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i), \underline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \right), \left(\overline{\mathcal{B}}_{h_{\mathcal{W}}}(\varsigma_i), \overline{\mathcal{U}}_{h_{\mathcal{W}}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \mathcal{T} \right\}$ and $\mathcal{X} = \left\{ \left\langle \varsigma_i, \left(\underline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i), \underline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \right), \left(\overline{\mathcal{B}}_{h_{\mathcal{X}}}(\varsigma_i), \overline{\mathcal{U}}_{h_{\mathcal{X}}}(\varsigma_i) \right) \right\rangle \mid \varsigma_i \in \mathcal{T} \right\}$ be in $\varrho = \{\varsigma_1, \varsigma_2, \varsigma_3, \dots, \varsigma_n\}$. The weighted cosine SM between two q-ROHFRSs $\mathcal{W}$ and $\mathcal{X}$ is defined as:

$$WC_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = \frac{1}{n} \bigoplus_{i=1}^n \omega_i \frac{\xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i) \xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i) + \xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i) \xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i)}{\sqrt{(\xi \mathcal{B}_{h_{\mathcal{W}}}(\varsigma_i))^2 + (\xi \mathcal{U}_{h_{\mathcal{W}}}(\varsigma_i))^2} \sqrt{(\xi \mathcal{B}_{h_{\mathcal{X}}}(\varsigma_i))^2 + (\xi \mathcal{U}_{h_{\mathcal{X}}}(\varsigma_i))^2}},$$

where $\omega_i \in [0, 1]$, $i = 1, 2, 3, \dots, n$ and $\bigoplus_{i=1}^n \omega_i = 1$.

Proposition 3.4 *Let $\mathcal{W}$, $\mathcal{X}$ and $\mathcal{Y}$ be three q-ROHFRSs on ϱ . The weighted cosine SM $WC_{q-ROHFRSs}(\mathcal{W}, \mathcal{X})$ is a mapping $S : q-ROHFRSs(\varrho) \times q-ROHFRSs(\varrho) \rightarrow [0, 1]$, carrying the following properties:*

- (A₁) $0 \leq C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) \leq 1$
- (A₂) $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = C_{q-ROHFRSs}(\mathcal{X}, \mathcal{W})$
- (A₃) $C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) = 1$ iff $\mathcal{W} = \mathcal{X}$
- (A₄) If $\mathcal{W} \subseteq \mathcal{X} \subseteq \mathcal{Y}$, then

$$C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{W}, \mathcal{X}) \text{ and } C_{q-ROHFRSs}(\mathcal{W}, \mathcal{Y}) \leq C_{q-ROHFRSs}(\mathcal{X}, \mathcal{Y}).$$

Proof The demonstration is similar to the proof of the Proposition 3.2. □

Practical applications

Case study I

This section will outline a DM procedure for choosing the optimal site for an electric vehicle charging station (EVCS), utilizing q-ROHFRSs and the innovative concept of similarity measures. In addition, we use the suggested technique to choose EVCS locations and perform a detailed sensitivity analysis to determine its practicality and reliability. We provide a concise overview of electric cars as a form of sustainable energy transportation. These cars have the potential to significantly minimize noise pollution, greenhouse gas emissions, hazardous pollutants, and fossil fuel consumption. As battery technology continues to advance and concerns about environmental pollution and energy constraints become more widespread, electric cars are experiencing a surge in worldwide popularity and rapid expansion. The transportation sector has emerged as the primary contributor to carbon dioxide emissions worldwide. The ongoing integration of electric vehicles (EVs) is expected to have a positive impact on reducing carbon dioxide emissions. Compared to conventional vehicles, EVs emit 54 percent less carbon dioxide per mile. In addition, EVs batteries have a conversion rate of 59 to 62 percent for vehicle movement, whereas gas-powered vehicles have conversion rates of 17 and 21 percent. EV drivers accumulate a charging cost of approximately \$1.2 per gallon, which is significantly lower than the price of gasoline. The yearly average operational expense for an electric vehicles stands at around \$485, markedly less than the \$1117 associated with a traditional vehicle. In summary, the reduced operational expenses associated with electric vehicles render them a compelling choice for individuals seeking to economize on transportation costs. Furthermore, electric vehicles present a more sustainable option compared to traditional automobiles, generating no tailpipe emissions and contributing to a decrease in greenhouse gas emissions. As the global landscape evolves towards sustainable modes of transport, electric vehicles are gaining prominence as a favored option. Furthermore, one of the key advantages of electrical transportation is its significant contribution to economic growth. In 2017, the United States depended on imports to cover 19 percent of its petroleum consumption, as reported by the U.S. Department of Energy. This notable percentage underscores the nation's persistent reliance on external oil sources to fuel its transportation, industrial sectors, and overall economic framework. By acknowledging the significance of these imports, policymakers and stakeholders can gain deeper insights into the ramifications for energy security, economic development, and environmental sustainability, thereby guiding strategies aimed at diminishing dependence on imported petroleum and facilitating a shift towards more domestic and renewable energy sources. The utilization of EVs has the potential to decrease our reliance on foreign energy sources and contribute to the growth of the economy through establishing employment opportunities, specifically in the skilled electrical trades. Based on the comprehensive growth strategy implemented by governmental bodies and manufacturers alike, the market share of EVs has the potential for substantial growth in the foreseeable future. However, the widespread use of electric cars will lead to substantial gaps in the development of charging infrastructure. Furthermore, the security and efficiency of the electricity grid could potentially face significant challenges. This section presents a numerical scenario for selecting an EVCS location.

The basic information of the computer and software used for simulations

- Windows: 8.1 single language
- Processor: Intel(R)Core(TM)i5-5200CPU@2.20GHz 2.20GHz
- Installed memory(RAM): 4.00GB

- System type: 64 bit Operating System, x64-based processor
- MATLAB
- Version: 7.10.0.499(R2010a)

The decision making process

This subsection illustrates the implementation of the suggested procedure for determining the most advantageous locations for EVCS utilising the q -ROHFRS approach. We evaluate three essential criteria for the selection of sites:

- Criteria-I Traffic convenience (ς_1): This criterion evaluates the convenience and accessibility of the EVCS location to significant transportation routes and demographic hubs.
- Criteria-II Construction cost (ς_2): This criterion assesses the financial expenses required to complete the establishment of the EVCS site.
- Criteria-III Service capability (ς_3): This criterion determines the capacity of the EVCS site to deliver effective and dependable charging services to users of electric vehicles.

A weight vector $w = [0.3, 0.5, 0.2]$ is assigned to these criteria, indicating their relative significance in the site selection process. A comprehensive description of the SMs based on q -ROHFRS and its implementation in the selection of EVCS sites is provided in the subsequent example:

Example 4.1 Let $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ be three site for EVCS and illustrated based on q -ROHFRSs is follows.

$$\begin{aligned}\mathcal{H}_1 &= \left(\begin{array}{l} \{\varsigma_1, \langle (0.30, 0.40, 0.60), (0.20, 0.40) \rangle, \langle (0.30, 0.90), (0.10, 0.20, 0.30) \rangle\}, \\ \{\varsigma_2, \langle (0.10, 0.40), (0.50, 0.70) \rangle, \langle (0.30, 0.40, 0.60), (0.60, 0.80, 0.90) \rangle\}, \\ \{\varsigma_3, \langle (0.30, 0.50), (0.50, 0.80) \rangle, \langle (0.30, 0.40, 0.60), (0.60, 0.90) \rangle\} \end{array} \right), \\ \mathcal{H}_2 &= \left(\begin{array}{l} \{\varsigma_1, \langle (0.10, 0.40, 0.70), (0.50, 0.60) \rangle, \langle (0.10, 0.50), (0.10, 0.20, 0.70) \rangle\}, \\ \{\varsigma_2, \langle (0.10, 0.40), (0.20, 0.70) \rangle, \langle (0.10, 0.40, 0.70), (0.20, 0.40, 0.50) \rangle\}, \\ \{\varsigma_3, \langle (0.20, 0.30), (0.30, 0.50) \rangle, \langle (0.30, 0.50, 0.70), (0.60, 0.80) \rangle\} \end{array} \right), \\ \mathcal{H}_3 &= \left(\begin{array}{l} \{\varsigma_1, \langle (0.40, 0.50, 0.70), (0.20, 0.90) \rangle, \langle (0.30, 0.70), (0.10, 0.50, 0.60) \rangle\}, \\ \{\varsigma_2, \langle (0.20, 0.40), (0.30, 0.70) \rangle, \langle (0.10, 0.40, 0.50), (0.70, 0.80, 0.90) \rangle\}, \\ \{\varsigma_3, \langle (0.10, 0.20), (0.20, 0.90) \rangle, \langle (0.30, 0.40, 0.60), (0.50, 0.60) \rangle\} \end{array} \right).\end{aligned}$$

The following is an optimal solution for EVCS site selection $\mathcal{G}$ evaluated in the form of q -ROHFRSs:

$$\mathcal{G} = \left(\begin{array}{l} \{\varsigma_1, \langle (0.10, 0.20, 0.30), (0.20, 0.30) \rangle, \langle (0.10, 0.30), (0.10, 0.30, 0.40) \rangle\}, \\ \{\varsigma_2, \langle (0.20, 0.40), (0.30, 0.40, 0.80) \rangle, \langle (0.10, 0.20, 0.30), (0.20, 0.30, 0.60) \rangle\}, \\ \{\varsigma_3, \langle (0.20, 0.50), (0.20, 0.70) \rangle, \langle (0.10, 0.30, 0.70), (0.30, 0.50) \rangle\} \end{array} \right).$$

The objective of a professional expert is to ascertain the optimal site for EVCS. To achieve this, the resemblance between $\mathcal{G}$ and the locations $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ is assessed, after which $\mathcal{G}$ is assigned to the designated site $\mathcal{H}_g$ as indicated by:

$$g = \arg \max_g \{D(\mathcal{H}_g, \mathcal{G})\}.$$

In the framework of the newly developed cosine SM ($C_{q-ROHFRSs}$) and weighted cosine SM ($WC_{q-ROHFRSs}$), the similarity measures between $C_{q-ROHFRSs}(\mathcal{H}_1, \mathcal{G})$, $C_{q-ROHFRSs}(\mathcal{H}_2, \mathcal{G})$, and $C_{q-ROHFRSs}(\mathcal{H}_3, \mathcal{G})$ are thoroughly determined and demonstrated in Table 1. The weight vector for $WC_{q-ROHFRSs}$ is considered as $w = [0.3, 0.5, 0.2]$. It has been observed in Table 1 that an unknown pattern $\mathcal{G}$ is classified within the category $\mathcal{H}_1$ when employing $C_{q-ROHFRSs}$ and $WC_{q-ROHFRSs}$. It is determined that the SMs between $C_{q-ROHFRSs}(\mathcal{H}_1, \mathcal{G})$ exceeds the SMs between $C_{q-ROHFRSs}(\mathcal{H}_3, \mathcal{G})$, which in consequently exceeds the SM between $C_{q-ROHFRSs}(\mathcal{H}_2, \mathcal{G})$. It seems that $C_{q-ROHFRSs}(\mathcal{H}_1, \mathcal{G})$ is the more favorable option. Through systematic computations, we may determine the previously identified relation for $WC_{q-ROHFRSs}$. The findings are presented in Table 1. It is evident that the optimal location is $\mathcal{H}_1$. It is important to observe that the computations obtained from the innovative SMs $C_{q-ROHFRSs}$ and $WC_{q-ROHFRSs}$ exhibit consistency with each other, yielding identical ranking orders. Figure 2 represents the ranking obtained from cosine and weighted cosine SMs.

Case study II

Let $\mathcal{H} = \mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3, \dots, \mathcal{H}_n$ denote the regions (distinct geographical regions), $\chi = \chi_1, \chi_2, \chi_3, \dots, \chi_n$ signify human activities (transport exhaust, smoking, airborne dust), while $\mathfrak{J} = \mathfrak{J}_1, \mathfrak{J}_2, \mathfrak{J}_3, \dots, \mathfrak{J}_n$ and

	$S_1(\mathcal{H}_1, \mathcal{G})$	$S_2(\mathcal{H}_2, \mathcal{G})$	$S_3(\mathcal{H}_3, \mathcal{G})$	Ranking	Classification results
$C_{q-ROHFRSs}$	0.2791	0.2563	0.2712	$S_1 > S_3 > S_2$	$\mathcal{H}_1$
$WC_{q-ROHFRSs}$	0.0922	0.0822	0.0905	$S_1 > S_3 > S_2$	$\mathcal{H}_1$

Table 1. Cosine weighted cosine SMs for Example.

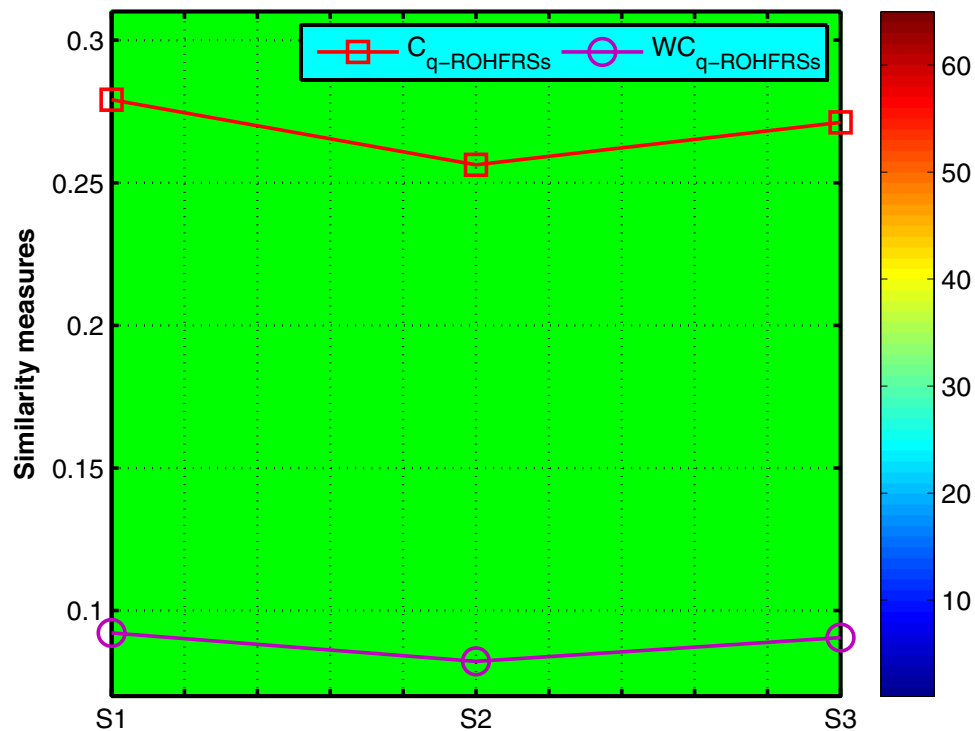


Fig. 2. The visual representation of the rankings obtained from cosine and weighted cosine SMs.

	$S_1(\mathcal{H}_1, \mathcal{G})$	$S_2(\mathcal{H}_2, \mathcal{G})$	$S_3(\mathcal{H}_3, \mathcal{G})$	Ranking	Classification results
$C_{q-ROHFRSs}$	0.4121	0.3851	0.0885	$S_1 > S_2 > S_3$	$\mathcal{H}_1$
$WC_{q-ROHFRSs}$	0.1333	0.1225	0.0296	$S_1 > S_2 > S_3$	$\mathcal{H}_1$

Table 2. Cosine and weighted cosine SMs for Example 4.2.

$\mathcal{T} = \mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3, \dots, \mathcal{T}_n$ represent the humidity levels (temperature, pressure) and air pollution (sulfur dioxide (SO_2), nitrogen (NO_2), carbon monoxide (CO), ozone (O_3)) in those particular regions, respectively. A decision-maker identifies the region and the factors $\chi, \mathcal{I}, \mathcal{T}$ that improve the Air Quality Index (AQI) by the application of a decision-making technique. This analysis can ascertain whether a geographical location exhibits favorable, acceptable, or hazardous AQI levels. This example presents the computation of the AQI for various cities with a data set derived from q -ROHFRSs with distinct pollution components. Utilizing the suggested similarity measures, we computed the values of the attributes and alternatives, as illustrated in form of q -ROHFRSs. The air quality index for each region is shown in Table 2, with its graphical representation in Fig. 3, indicating that the $\mathcal{H}_1$ region exhibits superior air quality relative to the $\mathcal{H}_2$ and $\mathcal{H}_3$ regions. The AQI of geographical region $\mathcal{H}_1$ is significantly elevated and may signify detrimental effects on human health. Our method for determining the air quality index of different regions becomes beneficial for motives of selection. The AQI of different regions can be readily computed using this mathematical method.

Example 4.2 Let $\mathcal{H}_1, \mathcal{H}_2$, and $\mathcal{H}_3$ be three site for geographical region and illustrated based on q -ROHFRSs as follows.

$$\begin{aligned}
 \mathcal{H}_1 &= \left(\begin{array}{l} \{s_1, \langle (0.40, 0.50, 0.60), (0.30, 0.50) \rangle, \langle (0.40, 0.90), (0.20, 0.30, 0.40) \rangle\}, \\ \{s_2, \langle (0.20, 0.50), (0.60, 0.80) \rangle, \langle (0.40, 0.50, 0.70), (0.70, 0.80, 0.90) \rangle\}, \\ \{s_3, \langle (0.40, 0.60), (0.60, 0.90) \rangle, \langle (0.40, 0.50, 0.70), (0.70, 0.90) \rangle\} \end{array} \right), \\
 \mathcal{H}_2 &= \left(\begin{array}{l} \{s_1, \langle (0.20, 0.50, 0.80), (0.60, 0.70) \rangle, \langle (0.20, 0.60), (0.20, 0.30, 0.80) \rangle\}, \\ \{s_2, \langle (0.20, 0.50), (0.20, 0.70) \rangle, \langle (0.10, 0.50, 0.80), (0.30, 0.50, 0.60) \rangle\}, \\ \{s_3, \langle (0.30, 0.40), (0.40, 0.60) \rangle, \langle (0.40, 0.60, 0.80), (0.70, 0.90) \rangle\} \end{array} \right), \\
 \mathcal{H}_3 &= \left(\begin{array}{l} \{s_1, \langle (0.50, 0.60, 0.80), (0.30, 0.90) \rangle, \langle (0.40, 0.80), (0.20, 0.60, 0.80) \rangle\}, \\ \{s_2, \langle (0.30, 0.50), (0.30, 0.70, 0.90) \rangle, \langle (0.30, 0.60, 0.80), (0.50, 0.70, 0.90) \rangle\}, \\ \{s_3, \langle (0.30, 0.50), (0.30, 0.90) \rangle, \langle (0.40, 0.60, 0.70), (0.60, 0.70) \rangle\} \end{array} \right).
 \end{aligned}$$

The following is an ideal geographical region with affective factors assessed using the framework of q -ROHFRSs:

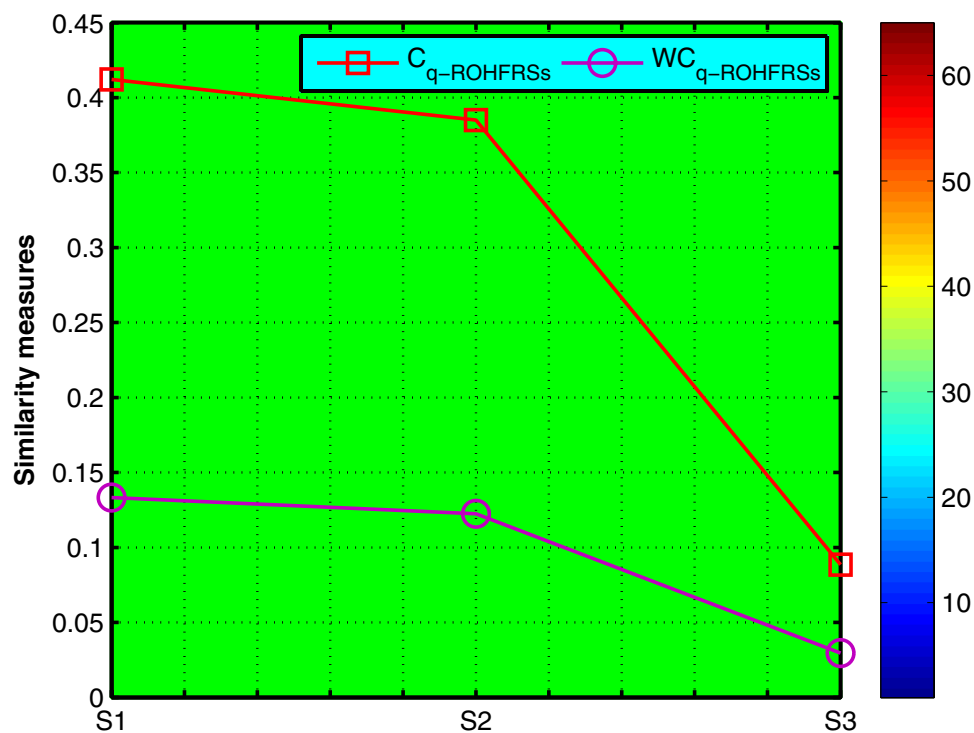


Fig. 3. The graphical depiction of the rankings determined by cosine and weighted cosine similarity measures.

Relation-1	Temperature	Headache	SP
p_1	$\left(\begin{array}{c} \left\{ (0.6, 0.0, 0), (0.3, 0, 0) \right\}, \\ \left\{ (0.8, 0, 0), (0.1, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.4, 0, 0), (0.4, 0, 0) \right\}, \\ \left\{ (0.6, 0, 0), (0.2, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.5, 0, 0), (0.2, 0, 0) \right\}, \\ \left\{ (0.7, 0, 0), (0.2, 0, 0) \right\} \end{array} \right)$
p_2	$\left(\begin{array}{c} \left\{ (0.5, 0, 0), (4.0, 0, 0) \right\}, \\ \left\{ (0.7, 0, 0), (0.2, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.5, 0, 0), (0.3, 0, 0) \right\}, \\ \left\{ (0.7, 0, 0), (0.3, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.5, 0, 0), (0.4, 0, 0) \right\}, \\ \left\{ (0.7, 0, 0), (0.4, 0, 0) \right\} \end{array} \right)$
p_3	$\left(\begin{array}{c} \left\{ (0.6, 0, 0), (0.4, 0, 0) \right\}, \\ \left\{ (0.8, 0, 0), (0.2, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.5, 0, 0), (0.3, 0, 0) \right\}, \\ \left\{ (0.7, 0, 0), (0.1, 0, 0) \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ (0.4, 0, 0), (0.4, 0, 0) \right\}, \\ \left\{ (0.8, 0, 0), (0.2, 0, 0) \right\} \end{array} \right)$

Table 3. (Relation-1) The relationship between patients and their symptoms.

$$\mathcal{G} = \left(\begin{array}{c} \{s_1, \langle (0.50, 0.80, 0.90), (0.40, 0.50) \rangle, \langle (0.50, 0.80), (0.30, 0.40, 0.60) \rangle\}, \\ \{s_2, \langle (0.30, 0.50), (0.30, 0.40, 0.80) \rangle, \langle (0.40, 0.60, 0.70), (0.40, 0.60, 0.70) \rangle\}, \\ \{s_3, \langle (0.50, 0.80), (0.50, 0.80) \rangle, \langle (0.60, 0.70, 0.90), (0.60, 0.90) \rangle\} \end{array} \right).$$

Validation of the proposed approach: a comparative study

To assess the validity and robustness of the proposed similarity measures, this section presents a comparative analysis between the results obtained in this study and those previously documented in the literature.

Example 5.1 In this example, we compared a medical diagnosis problem discussed by Pramanik et al.⁴⁹ and Attaullah et al.³⁵ for validation of the proposed approach. The medical diagnosis composed of uncertainties and maximum information available to physicians from new medical technologies. The act of classifying diverse set of symptoms under a mono disease is too arduous and challenging task. It can involve the medical diagnosis indulging more indeterminacy. After a comparison, we concluded that the findings aligned with those presented by Pramanik et al.⁴⁹ and Attaullah et al.³⁵. The proposed similarity measures among the patients versus symptoms and symptoms versus diseases provided the proper medical diagnosis. Let $P = \{p_1, p_2, p_3\}$, be a set of patients, $D = \{\text{Viral Fever (VF), Malaria, Stomach Problem (SP), Chest Problem (CP)}\}$, be a set of diseases and $S = \{\text{Temperature, Headache, Stomach Pain, Cough, Chest Pain}\}$ be a set of symptoms with the weight vector

Relation-1	Cough	CP
p_1	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.8, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.4, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$
p_2	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.9, 0, 0), \\ (0.3, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.7, 0, 0), \\ (0.3, 0, 0) \end{array} \right\} \end{array} \right)$
p_3	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.8, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.7, 0, 0), \\ (0.1, 0, 0) \end{array} \right\} \end{array} \right)$

Table 4. (Relation-1) The relation between patients and symptoms.

Relation-2	VF	Malaria
Temperature	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.8, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.1, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$
Headache	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.7, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.2, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$
SP	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.2, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.4, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.1, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.3, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$
Cough	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.4, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.6, 0, 0), \\ (0.1, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.3, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.5, 0, 0), \\ (0.3, 0, 0) \end{array} \right\} \end{array} \right)$
CP	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.2, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.4, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \end{array} \right)$	$\left(\begin{array}{c} \left\{ \begin{array}{c} (0.1, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \\ \left\{ \begin{array}{c} (0.3, 0, 0), \\ (0.1, 0, 0) \end{array} \right\} \end{array} \right)$

Table 5. (Relation-2) The relation among symptoms and diseases.

$\mathcal{W} = \{0.14, 0.16, 0.10, 0.20, 0.40\}$. The primary approach is to evaluate the patient and identify the disease presented in Tables 3, 4, 5, 6 as follows:

A comprehensive investigation of the data illustrated in Table 7 and Figs. 4, 5, 6 reveals a notable correlation between the similarity measures and precise diagnosis. Significantly, the most elevated measured consistently reflects the accurate diagnosis, thus validating the effectiveness of our suggested methodology. Our findings indicate that all three patients, p_1, p_2, p_3 , demonstrate the highest similarity measures for viral fever, thereby providing clear evidence that they are afflicted by this condition. The results are visually depicted in Figs. 4, 5, 6, presenting the outcomes of the similarity measures for each patient. Consequently, it could be observed that all the patients p_1, p_2, p_3 are afflicted with viral fever. The findings illustrate the efficacy of our methodology in attaining precise diagnoses and underscore the importance of similarity measures in determining the fundamental disease. The uniformity of the present results among all three patients highlights the dependability and strength of the novel approach, establishing it as a potentially valuable tool for medical diagnosis.

Example 5.2 ¹⁷ Let the three known patterns be denoted as $\mathcal{H}_1 (i = 1, 2, 3)$, are defined as follows:

$$\begin{aligned} \mathcal{H}_1 &= (\{ (0.35, 0.0), (0.35, 0.0), (0.20, 0.0), (0.47, 0.0) \}), \\ \mathcal{H}_2 &= (\{ (0.33, 0.0), (0.35, 0.0), (0.21, 0.0), (0.46, 0.0) \}), \\ \mathcal{H}_3 &= (\{ (0.31, 0.0), (0.31, 0.0), (0.20, 0.0), (0.47, 0.0) \}), \end{aligned}$$

and the unknown pattern $\mathcal{G}$ is defined as:

Relation-2	SP	CP
Temperature	$\left(\left\{ \begin{array}{l} (0.3, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.5, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \right)$	$\left(\left\{ \begin{array}{l} (0.2, 0, 0), \\ (0.6, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.4, 0, 0), \\ (0.4, 0, 0) \end{array} \right\} \right)$
Headache	$\left(\left\{ \begin{array}{l} (0.2, 0, 0), \\ (0.3, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.4, 0, 0), \\ (0.1, 0, 0) \end{array} \right\} \right)$	$\left(\left\{ \begin{array}{l} (0.1, 0, 0), \\ (0.5, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.5, 0, 0), \\ (0.3, 0, 0) \end{array} \right\} \right)$
SP	$\left(\left\{ \begin{array}{l} (0.4, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.6, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \right)$	$\left(\left\{ \begin{array}{l} (0.1, 0, 0), \\ (0.6, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.3, 0, 0), \\ (0.4, 0, 0) \end{array} \right\} \right)$
Cough	$\left(\left\{ \begin{array}{l} (0.1, 0, 0), \\ (0.6, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.3, 0, 0), \\ (0.4, 0, 0) \end{array} \right\} \right)$	$\left(\left\{ \begin{array}{l} (0.5, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.7, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \right)$
CP	$\left(\left\{ \begin{array}{l} (0.1, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.3, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \right)$	$\left(\left\{ \begin{array}{l} (0.4, 0, 0), \\ (0.4, 0, 0) \end{array} \right\}, \left\{ \begin{array}{l} (0.6, 0, 0), \\ (0.2, 0, 0) \end{array} \right\} \right)$

Table 6. (Relation-2) The correlation between symptoms and diseases.

	VF	Malaria	SP	CP
$C_{q\text{-ROHFRS}}(\text{proposed})$	–	–	–	–
p_1	0.5835	0.4963	0.5148	0.3510
p_2	0.6182	0.5421	0.5243	0.5072
p_3	0.5732	0.4284	0.4132	0.4273
$WC_{q\text{-ROHFRS}}(\text{proposed})$	–	–	–	–
p_1	0.1604	0.1523	0.1436	0.1343
p_2	0.1739	0.1658	0.1382	0.1236
p_3	0.1454	0.1327	0.1286	0.1346
Pramanik et al. ⁴⁹	–	–	–	–
p_1	0.9595	0.9114	0.8498	0.8743
p_2	0.9624	0.9320	0.8935	0.8307
p_3	0.9405	0.8873	0.8487	0.8372
Attaullah et al. ³⁵	–	–	–	–
p_1	0.4955	0.3871	0.4059	0.4400
p_2	0.5090	0.4321	0.4320	0.5083
p_3	0.4823	0.3965	0.4080	0.4562
Attaullah et al. ³⁵	–	–	–	–
p_1	0.0427	0.0133	0.0346	0.0306
p_2	0.0439	0.0369	0.0380	0.0426
p_3	0.0408	0.0327	0.0357	0.0320

Table 7. An analysis of the proposed similarity measures in comparison to the existing similarity measures.

$$\mathcal{G} = (\{ (0.37, 0.0), (0.31, 0.0), (0.23, 0.0), (0.44, 0.0) \}).$$

Our objective is to ascertain the classification of the unknown pattern $\mathcal{G}$. Consequently, we have calculated the degree of similarity between all established known patterns $\mathcal{H}_i (i = 1, 2, 3)$ and an unidentified pattern $\mathcal{G}$. Table 8 presents a comparison of the results obtained from the proposed method alongside those of existing approaches to demonstrate the efficacy and dependability of the results. The similarity values obtained from the proposed SMs, indicated by italic faces. Table 8 demonstrates that many established methods are insufficient for identifying the category of an unknown pattern. Furthermore, the SM proposed by Patel et al.⁶³ produced a similarity value of 1, while $\mathcal{H}_i \neq \mathcal{G}_i (i = 1, 2, 3)$. The proposed methods classify the unknown pattern as $\mathcal{H}_1$, thereby enhancing the reliability of the suggested SMs.

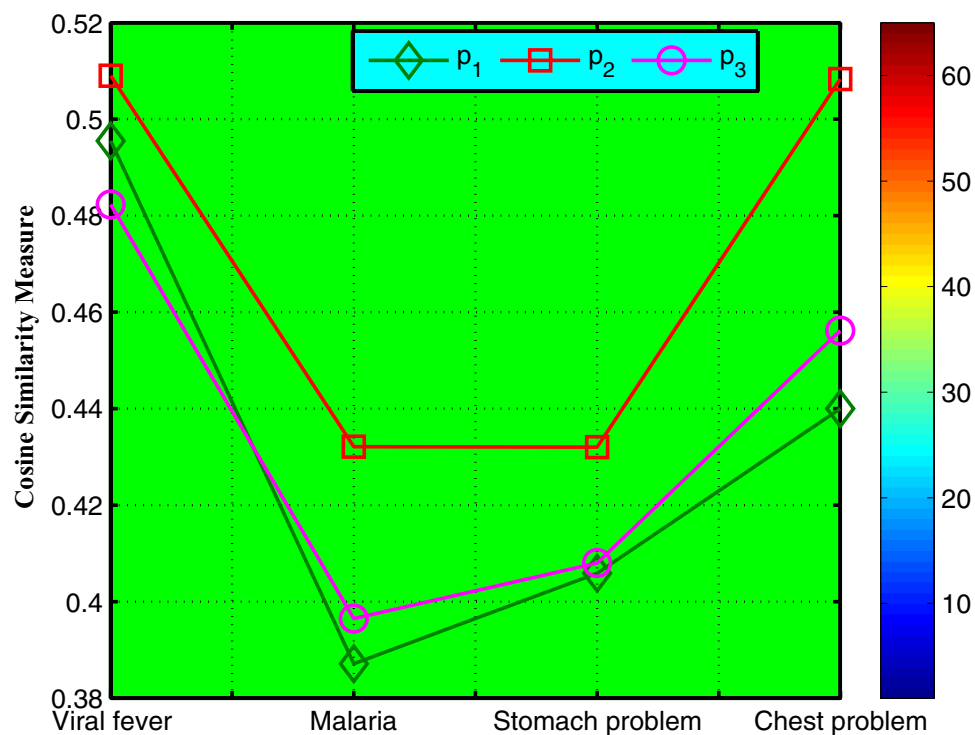


Fig. 4. The visual representation of the rankings obtained from the cosine SM.

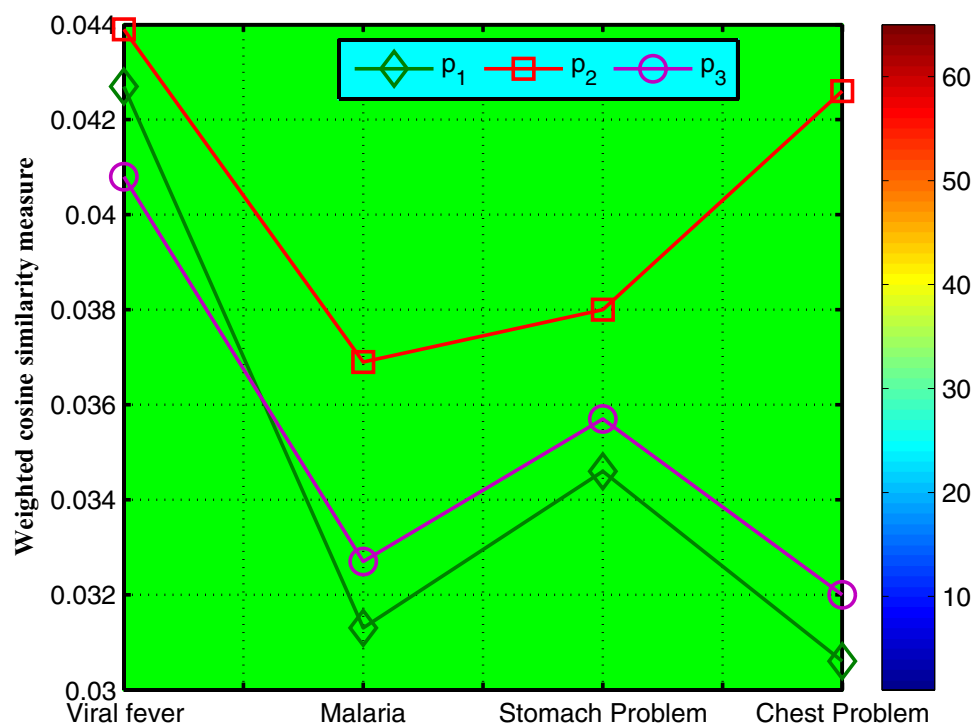


Fig. 5. The visual depiction of the ranking order established through weighted cosine SM.

The benefits of the proposed approach

The demonstrated SMs based on $q - ROHFRSs$ present significant benefits, interpreting these measures an essential resource for researchers and professionals by facilitating enhanced precision and efficacy in analysis and DM. The subsequent points delineate the primary benefits of the proposed SMs based on $q - ROHFRSs$.

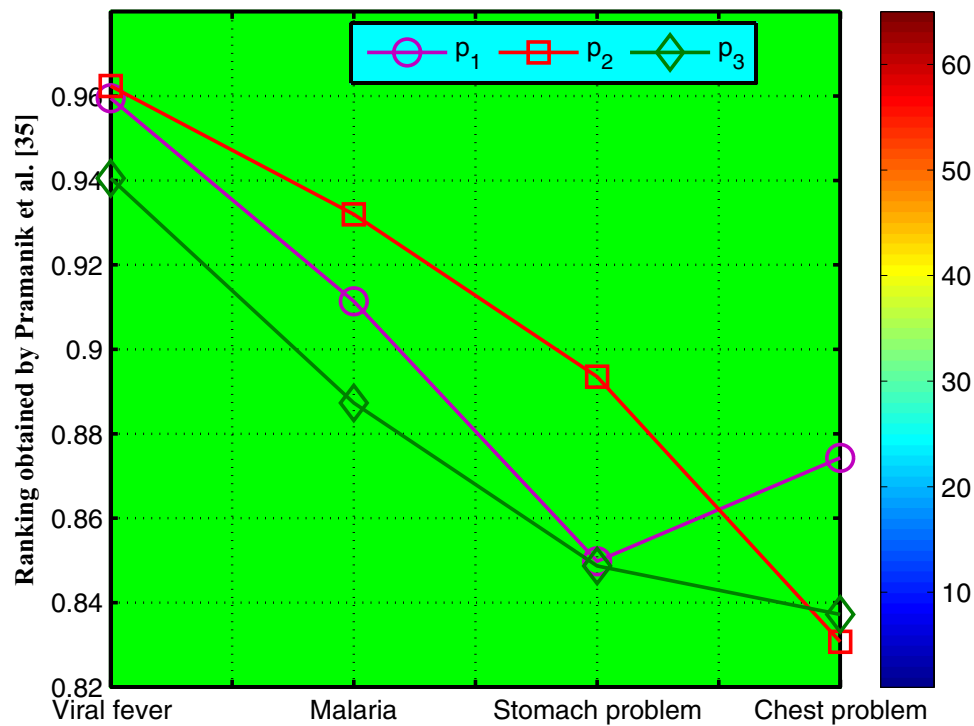


Fig. 6. The diagrammatic representation demonstrating the ranking achieved by Pramanik et al.⁴⁹.

References	$S(\mathcal{H}_1, \mathcal{G})$	$S(\mathcal{H}_2, \mathcal{G})$	$S(\mathcal{H}_3, \mathcal{G})$	Selection	Comment
Hong and Kim ⁵⁰	0.970	0.970	0.970	–	Failed to classify
Dengfeng and Chuntian ¹	.970	.968	.970	–	Failed to classify
Mitchell ⁵¹	.827	.827	.833	$\mathcal{H}_3$	Irrational
Wang and Xin ⁵²	.968	.970	.963	$\mathcal{H}_2$	Correctly classified
Vlachos and Sergiadis ⁵³	.997	.997	.996	–	Failed to classify
Hung and Yang ⁵⁴	.916	.916	.913	–	Failed to classify
Hung and Yang ⁵⁴	.970	.970	.970	–	Failed to classify
Hung and Yang ⁵⁴	.458	.457	.457	$\mathcal{H}_1$	Irrational
Hung and Yang ⁵⁴	.956	.956	.955	–	Failed to classify
Yang and Chiclana ⁵⁵	.974	.976	.965	$\mathcal{H}_2$	Correctly classified
Ye ⁴⁸	.996	.996	.996	–	Failed to classify
Boran and Akay ⁵⁶	.900	.900	.900	–	Failed to classify
Ye ⁵⁷	.999	.999	.997	–	Failed to classify
Ye ⁵⁷	.999	.999	.997	–	Failed to classify
Xiao ⁵⁸	.971	.973	.959	$\mathcal{H}_2$	Correctly classified
Mahanta and Panda ⁵⁹	.956	.956	.955	–	Failed to classify
Patel et al. ⁶⁰	.999	.999	.998	–	Failed to classify
Gohain et al. ⁶¹	.969	.969	.969	–	Failed to classify
Zeng et al. ⁶²	.987	.988	.987	$\mathcal{H}_2$	Correctly classified
Patel et al. ⁶³	1.000	1.000	1.000	–	Failed to classify
Patel et al. ¹⁷	.957	.964	.951	$\mathcal{H}_2$	Correctly classified
Jiang et al. ⁶⁴	.979	.979	.979	–	Failed to classify
Dutta et al. ⁶⁵	.927	.928	.928	–	Failed to classify
$C_q-ROHFRS(proposed)$	0.03238	0.03224	0.03138	$\mathcal{H}_1$	Correctly classified
$WC_q-ROHFRS(proposed)$	0.00971	0.00967	0.00941	$\mathcal{H}_1$	Correctly classified

Table 8. An evaluation of the suggested SMs in comparison with existing SMs models for Example 5.2.

1. The q – ROHFRSs model is a beneficial, practical, and astonishing enhanced form of fuzzy rough sets. In this instance, professional experts are more able to express their perspectives on the level of membership.
2. Whenever uncertain data are required to implement the constrained form of intuitionistic fuzzy numbers and Pythagorean fuzzy numbers, the identification of the best option from a group of alternatives in an MADM situation is hindered. The aforementioned possibilities would lead to information manipulation. To assure convincing responses in such critical instances, a more generalised model is necessary. As an effective extension of fuzzy rough sets, q -ROHFRSs provide more accurate and precise results when utilised to deal with real-world MADM challenges including q -ROHFR information.
3. The measures used in this research are not restricted. Working with both measures provides a geometric as well as an algebraic perspective on the MCDM challenge.

Conclusions and future scopes

The q -ROHFRSs is an incredibly valuable tool for expressing uncertain and ambiguous information. The challenge of how to precisely measure the similarity between two q -ROHFRSs remains unresolved. This research demonstrated the effectiveness of the cosine and weighted cosine SMs for q -ROHFRSs in addressing imprecise and ambiguous information. We developed two new types of cosine and weighted cosine SMs and investigated their relevant features in detail. The development of cosine and weighted cosine SMs has provided a robust framework for evaluating the similarity between q -ROHFRSs. To highlight the usefulness of the developed SMs, we presented a numerical example involving the identification of an optimal location for EVCS. The numerical example illustrating the optimal location selection for EVCS has validated the feasibility and reliability of the described approaches. Furthermore, an extensive experimental analysis for air quality index (AQI) was carried out to validate the superiority and effectiveness of the proposed approach that represents a crucial environmental parameter that is affected by the presence of air pollution. Air pollution represents a critical global challenge and is broadly acknowledged as detrimental to human well-being. The proposed SMs serve to evaluate the air quality index (AQI) of a specific geographical region, classifying it as good, moderate, or hazardous. The results and ranking order of both the suggested SMs confirmed the portability and robustness of the innovative approaches. Alongside emphasizing the findings, the applications illustrate the crucial role and trustworthiness of the established SMs. In order to explore the efficacy and precision of the proposed SMs, we conducted a comparative analysis of their outcomes contrary to those documented in the existing literature, utilizing improved cosine and weighted cosine techniques. The findings from the investigations suggest that the novel techniques of SMs could yield reliable classification results. The significance of this contribution may be measured at the industrial and academic as decision analysis and uncertainty theory and decision models. The concepts expressed and the literature explored may embed innovative notions. As a consequence, researchers and practitioners from all over the world will be attracted to flesh out and improve on the proposed framework. The application of q -ROHFRSs provides us with enhanced accuracy, though at the cost of significant computation time, resulting in a notable limitation. To overcome this challenge, we are developing Python and MATLAB codes.

Future research directions will focus on:

- Investigating the applications of the proposed similarity measure in various decision-making domains, including bidirectional approximate reasoning, clustering, and image segmentation.
- Establishing innovative entropy measures based on the proposed similarity framework and assessing their effectiveness in practical decision-making contexts.
- Expanding the proposed work to other extensions of the FS framework to improve accuracy and computational efficiency. This study contributes significantly to the domains of fuzzy sets and rough sets, presenting a novel perspective on SMs for q -ROHFRSs. The proposed approaches have shown promising results, and future research will investigate their potential across various applications^{66–69}.

Data availability

All the data are available in the manuscript.

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Author contributions

All the authors contributed equally to the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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